9 **(a)** **(i)** Explain what is meant by the binding energy of a nucleus.

It is the energy required to separate the nucleons in a nucleus to infinity. [1]

The variation of the binding energy per nucleon with nucleon number of all the different nuclei is shown in Fig. 9.1.

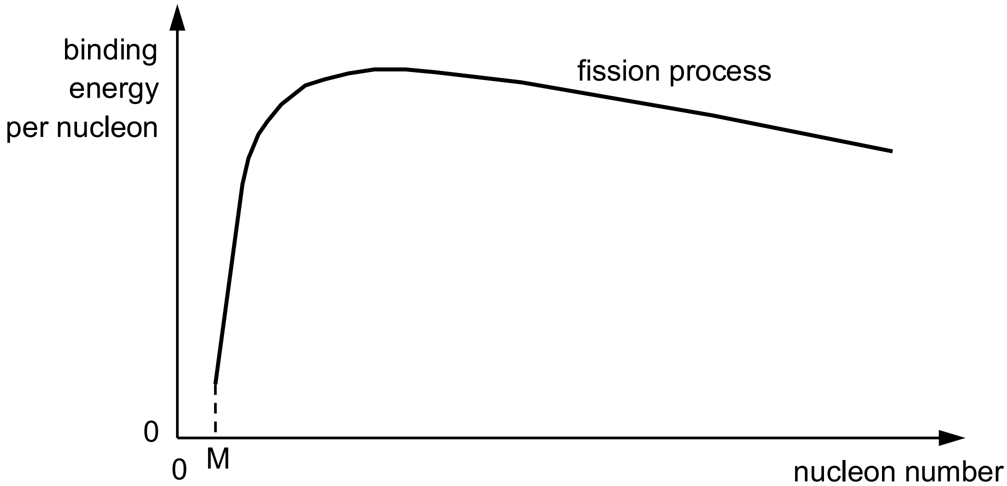

Fig. 9.1

(ii) State the value of M and explain your answer clearly.

Value of M is 2. If the nucleon number is only 1, it cannot be separated further. Hence the minimum number of nucleons needed to form a nucleus is 2.

[2]

(b) In a fission process, a neutron with speed of 2.00×10^3 m s^{-1} collides with a Uranium-235 nucleus and causes a nuclear reaction summarised in the following equation.

$$ {}^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{236}_{92}U \rightarrow {}^{143}_{54}Xe + {}^{90}_{38}Sr + 3\,{}^{1}_{0}n + \text{energy} $$

(i) Using the data from Fig. 9.2, show that the binding energy per nucleon for Strontium-90 is 8.73 MeV.

	rest mass
Strontium, ${}^{90}_{38}Sr$	89.9077 u
Proton, ${}^{1}_{1}p$	1.0078 u
Neutron, ${}^{1}_{0}n$	1.0087 u

Fig. 9.2

Mass defect = $(38m_p + 52m_n) - m_{nucleus}$

= (38)(1.0078u) + (90-38)(1.0087u) – 89.9077 u = 0.8411 u.

BE = mass defect $\times$ c^2 = $(0.8411 \times 1.66 \times 10^{-27})(3.00 \times 10^8)^2(1/1.60 \times 10^{-19})$ eV

= 785 MeV

Hence BE per nucleon = BE/n = 8.73 MeV

[3]

(ii) Hence calculate the total energy released during the nuclear fission reaction. Data for binding energies per nucleon are shown in Fig. 9.3.

Isotope	binding energy per nucleon / MeV
Uranium-235	7.59
Xenon-143	8.41

Energy released = (total B.E.)$_{products}$ - (total B.E.)$_{reactants}$

= $[(90)(8.73) + (143)(8.41) - (235)(7.59)] \times 10^6 \times 1.60 \times 10^{-19}$ J

= 3.27×10^{-11} J

energy released = ... J [2]

(iii) Explain quantitatively why the kinetic energy of the neutron directed at Uranium-235 is neglected.

KE of neutron = $(1/2mv^2) = (1/2)(1.0087 \times 1.66 \times 10^{-27})(2.00 \times 10^3)^2$

= 3.35×10^{-21} J

Since KE of neutron << energy released, it is neglected.

.... [1]

(c) Fig. 9.4 shows the possible directions of Strontium-90, Xenon-143, and neutrons when nuclear fission takes place. Assume that this is an isolated system.

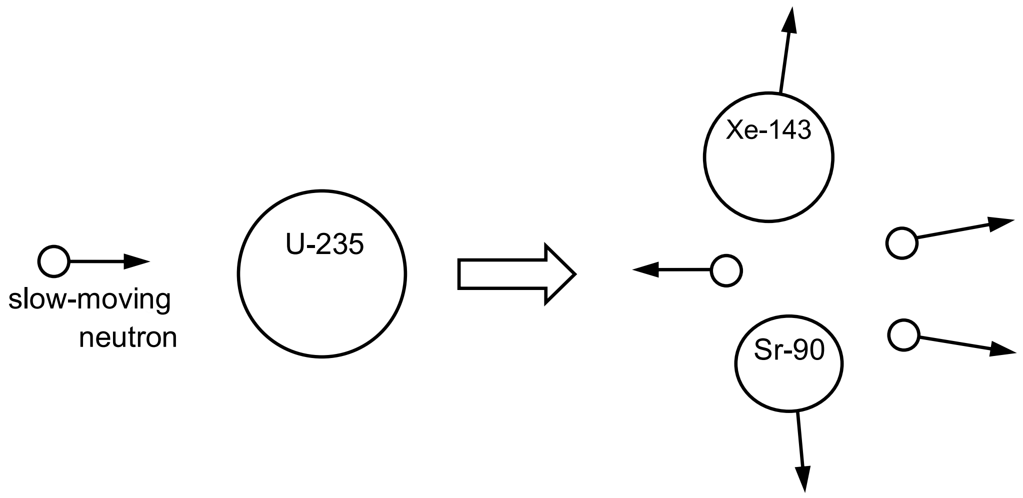

Fig. 9.4

(i) Explain why it is unlikely for the two fission products to move in the same direction after the fission process.

The total initial momentum of the reactants is close to zero as the neutron is slow moving. By principle of conservation of linear momentum, the total momentum of the products should be close to zero as well. Since the two fission products are much more massive than the neutrons released, the neutrons will have to travel approximately in the opposite direction of the two fission products with speeds much larger than them such that the total momentum is approximately zero. This may not be energetically possible and it is unlikely for the two fission products to move in the same direction. [1]

(ii) The total momentum of two fission products and neutrons after reaction is not zero even though the total momentum of Uranium-235 and slow moving neutron may be taken to be zero. Explain why this may be so.

Part of the energy released may be due to photon such as gamma radiation. The momentum of photon may need to be taken into consideration since here is a emission of gamma photon. The photon will move off in a direction such that the vector sum of the total final momentum will be zero. [2]

(d) Neutrons emitted from a nuclear fission may hit another Uranium-235 nucleus, causing a chain reaction.

(i) To control the reaction, neutrons emitted from a fission reaction may be slowed down by Carbon-12 nuclei, ${}^{12}_{6}C$. Show that when a neutron undergoes an elastic head-on collision with a Carbon-12 nucleus as shown in Fig. 9.5, the speed is reduced by about 15%.

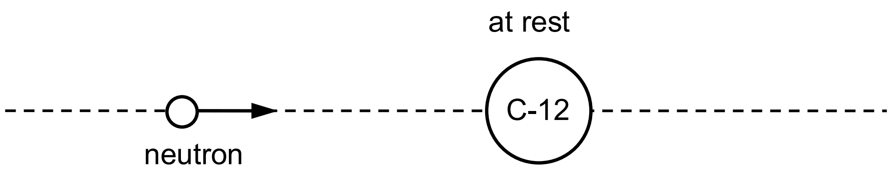

Fig. 9.5

$\Sigma p_{inital} = \Sigma p_{final}$

$(m)(u) = m(v_1) + (12m)(v_2)$

apply relative speed equation

$v_2 - v_1 = u$

Solving

$v_1 = -0.85\ u$

[3]

(ii) On average, the neutron speed after each collision is 0.93 of its speed before the collision.

Suggest why this speed reduction is different from what is stated in **(ii)** for a nuclear reactor.

The collision is unlikely to be always head-on and hence the kinetic energy transferred to C-12 will be less. Hence the speed reduction will be lesser.

[1]

(iii) Suggest why a slow neutron has a higher chance of being captured by U-235 to cause a fission reaction compared to a fast neutron.

The probability that fission will occur is described by the neutron cross-section. The fission cross-section increases greatly when the speed reduces making the likelihood of interaction with U-235 greater.

[1]

(e) The fission products are usually radioactive and give rise to a series of radioactive decay products. Each decay product has its own half life, but eventually a stable nuclide is reached.

One such fission product and its decay products is shown here:

$${}^{143}_{54}Xe \longrightarrow {}^{143}_{55}Cs \longrightarrow {}^{143}_{56}Ba$$

The half-life of Xe-143 and Cs-143 are 0.511 s and 1.79 s respectively.

(i) Suggest how the number of Cs-143 nuclei inside the nuclear reactor may remain constant even when it decays to form Ba-143.

Cs-143 is continuously formed from the decay of Xe-142. The number of Cs-143 may remain constant if the rate of decay of Xe is equal to the rate of decay of Cs.

[2]

(ii) Explain why a particular Xenon-143 nucleus produced earlier in the reaction may not necessarily decay before another Xenon-143 nucleus which was produced after it.

Due to the **random** nature of radioactivity decay, the probability of decay of a nucleus (per unit time) is constant (and the same for all radioactive Xe-143 nuclei). Hence, there is possibility that Xenon produced at a later time may decay first.

[1]

[Total: 20]